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Imperceptible adversarial audio steganography based on psychoacoustic model

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Abstract

Recently, deep learning based audio steganalysis methods have demonstrated superior performance in detecting the conventional audio steganography, which poses great chanllegnes to the conveiontional audio steganography. In this work, observed that the neural network can easily be deceived by specially perturbed inputs, i.e., adversarial examples, we propose an imperceptible audio steganography method based on psychoacoustic model. Specifically, we first add perturbation on the stego audio for constructing noise stego audio, which is delivered to the trained steganalyzer for misclassification. The perturbation is optimized in the adversarial process, aiming to seek an optimal perturbation that guarantee the imperceptibility and undetectability of stego audio. Further consider that the difficulty to optimize the threshold loss function using gradient back-progagation, we adopt two-stage optimization strategy to minimize the loss function. The first stage attempts to find a suitable perturbation to deceive the steganalyzer. The second stage concentrates on further optimizing the perturbation to make the stego imperceptible. For the practical steganography, the optimal perturbation obtained from the adversarial attack process is added on the original cover audio to construct the adversarial cover audio. Then one can use information embedding algorithm to embed the secret message on the adversarial cover to generate stego audio. Extensive experiments show that the proposed method can generate the adversarial cover audio with high perceptual quality and the undetectability performance outperforms the conventional audio steganography schemes.

Keywords Audio steganography · Adversarial examples · Psychoacoustic model · Masking threshold

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1 Introduction

Digital steganography is the art and science of utilizing the human perception redundancy to embed secret message into a digital cover such as video, image and audio. The cover with embedded data, i.e., the stego, could bypass adversary monitoring and realize the covert communication. The steganogrpahy research has achieved significant progress in the past decades.

On the grounds of different embedding strategies, steganography can be divided into non-adaptive steganography and adaptive steganography. Generally, non-adaptive steganography usually modifies all elements of cover in an indiscriminate manner. The typical works following this guideline are LSB [1] and LSB Matching (LSBM) [23]. Instead, adaptive steganography considers the attributes of cover and selectively embeds secret messages in areas with complex textures. The reputational adaptive steganogrpahy algorithms include WOW [9], HUGO [20], HILL [13], S-UNIWARD [10]. These methods are all implemented on Syndrome-Trellis Codes (STC) [4]. In the adaptive steganography framework, a distortion cost function of each embedding position in the cover is defined to characterize the degree of distortion for stego, then the total distortion is minimized under the assumption that the embedded operations are independent of each other. It should be noted that the adaptive steganography can be easily detected by Spatial Rich Model (SRM) [5] steganalysis method. This method elaborately designs multiple high-pass filters to preprocess the stego to magnify steganographic signals, then ensemble classifier is employed to detect stego with superior classification performance.

In the recent years, deep learning has achieved great breakthroughs in computer vision, natural language processing, speech recognition and so on, and it has infused the fresh vitality into the information hiding domain in the last few years. While the deep learning based steganalysis methods [29, 30] have significantly outperformed the traditional isteganalysis methods [11, 19] in detecting conventional steganography algorithms, which may challenge the development of steganography algorithms. Researchers combine the peculiarities of digital steganography to propose deep learning based steganogrpahy algorithms for withstanding the deep learning based steganalysis methods. Volkhonskiy et al. [28] first proposed a generative steganogaphy model Steganographic Generative Adversarial Network (SGAN) to generate cover images with better embedding properties. Unlike SGAN, Hayes and Danezisv [8] proposed to use the secret message and cover image to generate secret message. Tang et al. [27] combined GAN with adaptive steganography to propose automatic steganographic distortion learning framework with GAN (ASDL-GAN). This steganography method could automatically learn the distortion cost. Tang et al. [25] creatively proposed the SPAR-RL steganography model based on Deep Reinforcement Learning.

In addition to the GAN-based steganography approaches, researchers also proposed several steganography schemes based on adversarial examples. Zhang et al. [31] adopted Fast Gradient Sign Method (FGSM) [7] to design a steganography model, of which the core idea is to add random noise for simulating embedding operation on the cover image to generate noise stego image, and then conducted adversarial attack on the deep learning based steganalyzer to yield perturbation. Finally the adaptive steganography algorithm was used to embed secret messages. The reported results illustrated that this method achieved excellent performance in resisting deep learning based steganalysis methods. Tang et al. [26] suggested that adversarial examples could be used to adjust steganographic distortion cost. This method employed the adversarial examples to generate stego images with minimum modification to attack the targeted steganalyzer.

However, most of the aforementioned works focus primarily on digital images while less work is investigated on the audio cover. The existing conventional audio steganograhpy cannot resist the deep learning based steganalysis methods. Lin-Net [15] and Chen-Net [3] are the two state-of-the-art deep learning based audio steganalysis methods, which have illustrated superior classification performance in detecting the conventional audio steganography methods. Recent studies have shown that most CNN-based classifiers can be easily deceived by adversarial examples, which yield slightly perturbed inputs to a target network via the gradients. Inspired from this phenomenon, we have proposed an imperceptible adversarial audio steganography with psychoacoustic model. In the optimization process, we first add on the stego audio for constructing noise stego audio that is delivered to the trained steganalyzer for misclassifying. The perturbation is constantly optimized in the adversarial process, this is aimed at seeking for an optimal perturbation that guarantee the imperceptibility and undetectability of stego audio. Empirically speaking, it's difficult to utilize the gradient back-propagation to optimize the threshold loss function directly because of its higher order in magnitude. So we adopt two-stage optimization strategy to minimize the loss function, the first stage aims to find a suitable perturbation to deceive the steganalyzer, the second stage concentrates on further optimizing the perturbation to make the stego imperceptible. For the practical steganography, the optimal perturbation obtained from the adversarial attack process is added on the original cover audio to construct the adversarial cover audio. Then one can use information embedding method (e.g., LSBM) to embed the secret message on the adversarial cover to generate stego audio. It should be remarked that the secret message can be correctly extracted as usual. Extensive experiments show that our proposed method can form the adversarial cover audio with high perceptual quality, and the undetectability performance significantly outperforms the classical audio steganography schemes. The main contributions of this paper are summarized as follows:

- We introduce psychoacoustic model to perform the audio steganography from the perspective of adversarial examples. This can make the stego audio more imperceptible.
- The elaborately designed objective function is composed of steganalysis loss and mask threshold loss, of which optimization is divided into two stages. This can make it easy to utilize the gradient to optimize the threshold loss function directly.
- Extensive experiments show that our proposed method can form the adversarial cover audio with high perceptual quality, and the undetectability performance significantly outperforms the classical audio steganography schemes.

2 Related work

2.1 Adversarial examples

Szegedy et al. [24] first introduced the concept of adversarial example, which refers to the phenomenon that the neural network could be easily fooled by the slightly perturbed inputs. Goodfellow et al. [7] proposed a simple adversarial example generation methods, i.e., FGSM. Alexey et al. [12] proposed an iterative adversarial sample generation method (IFGSM). The core idea was to optimize a large-step operation to enlarge the loss function of the classifier by processing multiple small steps to increase the loss function to perform image perturbation. This was to generate the superior adversarial example. Mardry et al. [18] proposed a more generalized version with the iterative gradient based l_∞ attack, i.e., Projected Gradient Descent (PGD).

2.2 The preliminary of psychoacoustic model

Psychoacoustics is the science of studying the statistical relationships between acoustic signals and hearing sensations. A good understanding of the human auditory systems (HAS) is of significant importance to form imperceptible audio adversarial examples. We have introduced the frequency masking in this work. Frequency masking refers to the phenomenon that the louder audio signal (the "masker") can shield other audio signals at nearby frequencies (the "maskees") [2, 14]. In the intuitive auditory experience, the maskees cannot be perceived by human auditory. To put it simply, the masker may be regarded as forming a "masking threshold" in the frequency domain. This makes any audio signals under the threshold are imperceptible.

In order to quantitatively characterize the masking threshold, the corresponding theoretical foundations should be given. The Short-Time Fourier Transform (STFT) of the original audio signal is first computed to acquire the spectrum of overlapping sections (the "window") of one signal. The window size N is set to 2048, and the "hop size" is 512. Then the log-magnitude Power Spectral Sensity (PSD) can be calculated by

$$p_x(k) = 10\log_{10}|\frac{1}{N}s_x(k)|^2 \tag{1}$$

where $s_x(k)$ denotes the k bin of the spectrum of frame x. Lin et al. [18] adopted the normalized PSD estimate $\overline{p_x(k)}$ to compute the masking threshold, and the normalized PSD estimate can be expressed as

$$\overline{p_x(k)} = 96 - \max_k\{p_x(k)\} + p_x(k) \tag{2}$$

The specific computation of the masking threshold $t_x(k)$ is given in Ref 24, we just introduce the masking threshold to design the objective function. When the perturbation is added on the audio input x, if the normalized PSD estimate of the perturbation $\overline{p_\delta(k)}$ is lower than the frequency masking threshold of the original audio $t_x(k)$, then the perturbation will be masked by the original audio. Hence the perturbation cannot be perceived by the human auditory. Correspondingly, the normalized PSD estimate of the perturbation $\overline{p_\delta(k)}$ is calculated by

$$\overline{p_\delta(k)} = 96 - \max_k\{p_x(k)\} + p_\delta(k) \tag{3}$$

where $p_\delta(k) = 10\log_{10}|\frac{1}{N}s_\delta(k)|^2$, $p_x(k) = 10\log_{10}|\frac{1}{N}s_x(k)|^2$.

3 Imperceptible adversarial cover for audio steganography

3.1 Overall framework

The adversarial attack flowchart of proposed audio steganography is illustrated in Fig. 1. Firstly, the information embedding algorithm, e.g., LSBM, is employed to embed the secret message into the cover, then perturbation is added on the stego audio for constructing noise stego audio that is delivered to the trained steganalyzer for misclassifying. The perturbation is constantly optimized in the adversarial process, this is aimed at seeking for an optimal perturbation that guarantee the imperceptibility and undetectability of stego audio. Empirically speaking, it's difficult to utilize the gradient back-propagation to optimize the threshold loss function directly because of its higher order in magnitude. So we adopt two-stage optimization strategy to minimize the loss function, the first stage aims to find a suitable

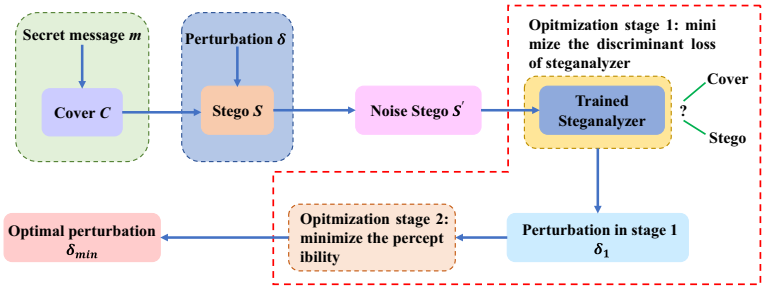


Fig. 1 The adversarial attack flowchart of proposed audio steganography

perturbation to deceive the steganalyzer, the second stage concentrates on further optimizing the perturbation to make the stego imperceptible.

Figure 2 shows the implementation process of practical steganography. The optimal perturbation $\delta_{\min}$ obtained from the adversarial attack process is added on the original cover audio to construct the adversarial one. Then one can use information embedding method (e.g., LSBM) to embed the secret message on the adversarial cover to generate stego audio. It should be remarked that the secret message can be correctly extracted as usual.

3.2 Objective function

The core of a steganalysis model is binary classifier, its decision criterion on the input s can be expressed as

$$\mathcal{F}(s) = \begin{cases} 0, & \text{if} \quad \Phi(s) < 0.5, \\ 1. & \text{if} \quad \Phi(s) \geq 0.5. \end{cases} \tag{4}$$

where $\mathcal{F}(\cdot)$ represents the deep learning based steganalyzer, $\Phi(\mathrm{s}) \in [0, 1]$ is decision probability of input s to manifest s belongs to a cover ($\mathcal{F}(s) = 1$) or a stego ($\mathcal{F}(s) = 0$).

Therefore, to make the adversarial cover audio enjoy high perceptual quality, we design the objective function elaborately to find an optimal perturbation in the adversarial steganography task. The objective function is expressed as

$$\mathcal{L} = \mathcal{L}_{\text{snet}} + \alpha \mathcal{L}_{\text{m}}(s + \delta, \delta) \tag{5}$$

where s represents the stego audio, δ denotes the optimizing perturbation, $\mathcal{L}_{\text{snet}}(\cdot)$ represents the steganalysis loss, $\mathcal{L}_{\text{m}}(\cdot)$ denotes the masking threshold loss, the hyperparameters

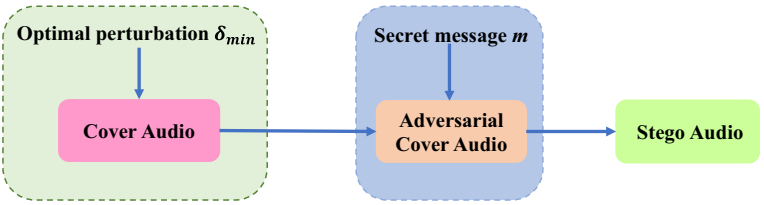


Fig. 2 The practical audio steganography scheme

α is used to balance the importance between this two losses. More specifically, $\mathcal{L}_{\text{snet}}(\cdot)$ is computed in terms of binary cross entropy, which can be calculated by

$$\mathcal{L}_{\text{snet}} = \mathbb{E}_{\delta}[\log(1 - \mathcal{F}(s + \delta))] \tag{6}$$

where $\mathbb{E}_{\delta}[\cdot]$ denotes the expectation operator over the input noise stego audio.

The masking threshold loss aims to optimize the first stage's perturbation for guaranteeing stego audio imperceptible. This can be expressed as

$$\mathcal{L}_{\text{m}}(s', \delta) = \frac{1}{\lfloor \frac{N}{2} \rfloor + 1} \sum_{k=0}^{\lfloor \frac{N}{2} \rfloor} \max\{\overline{p_{\delta}(k)} - t_{s'}(k), 0\} \tag{7}$$

where $\overline{p_{\delta}(k)}$ represents the normalized power spectral density of the perturbation δ.

3.3 Optimization strategy

The optimization process of objective function is divided into two stages. The hyperparameters α is set to 0 in the objective function when the optimization process lies in the first stage. This means that only steganalysis loss is used to optimize the perturbation for deceiving the steganalyzer. This process can be formulated as

$$\begin{aligned} &\text{argmin}_{\delta} \quad \mathbb{E}_{\delta}[\log(1 - \mathcal{F}(s + \delta))] \\ &s.t. \quad ||\delta|| < \epsilon \end{aligned} \tag{8}$$

where $||\delta||$ represents the max-norm ($||\cdot||_{\infty}$) of δ. The initial value of perturbation $||\delta||$ is 0, and δ is optimized by gradient descent algorithm in each epoch:

$$\delta = \delta - l_{r1} \cdot \text{sign}(\nabla_{\delta} \mathbb{E}_{\delta}[\log(1 - \mathcal{F}(s + \delta))]) \tag{9}$$

$$\text{Clip}||\delta|| \leq \epsilon \tag{10}$$

where l_{r1} represents the learning rate in stage 1, $\nabla_{\delta}\mathbb{E}_{\delta}[\cdot]$ denotes the steganalysis loss of gradient with respect to perturbation δ. The hyperparameter ϵ is the intensity of perturbation. We clip the perturbation in each epoch to prevent it from being too large.

When the optimization process lies in the second stage, we set the initial value of hyperparameters α to 0.06, which is dynamically updated on the basis of steganalysis attack performance. It means that the steganalysis loss and masking threshold loss both join for guiding the optimization of perturbation δ_1 obtained from stage 1. This stage concentrates on further optimizing the perturbation to make the stego imperceptible. For the same reason, δ_1 is also optimized by gradient descent algorithm in each epoch:

$$\delta = \delta - l_{r2} \cdot \nabla_{\delta}(\mathbb{E}_{\delta}[\log(1 - \mathcal{F}(s + \delta))] + \alpha \mathcal{L}_{\text{m}}((s + \delta), \delta)) \tag{11}$$

where l_{r2} represents the learning rate in stage 2. The hyperparameter α balance the importance between steganalysis loss and masking threshold loss. For the updating of α, if the optimization process run every 10 epochs, the perturbation can successfully attack the steganalyzer, we increase α to make the stego more imperceptible. If every 20 epochs, the perturbation cannot successfully attack the steganalyzer, α is decreased. After adequate adversarial attack, the optimal perterbation δ_{min} will be obtained. The details of optimization strategy is described in Algorithm 1.

Input: stego audio s, trained steganalyzer $\mathcal{F}$, perturbation δ, steganalysis loss $\mathcal{L}_{\text{snet}}$, masking threshold loss $\mathcal{L}_{\text{m}}$, hyperparameter ϵ and α, learning rate for the first stage l_{r1} and second stage l_{r2}, epoch of iterations in the first stage N_1 and second stage N_2.

Output: The optimal perturbation $\delta_{\min}$.

Stage 1: only steganalysis loss joins for optimizing the loss function.

Initialization: $\delta = 0, \epsilon = 1, l_{r1} = 0.003, l_{r2} = 0.0003, \alpha = 0$

for i = 0 to N_1-1 **do**

  $\delta = \delta - l_{r1} \cdot \text{sign}(\nabla_\delta \mathbb{E}_\delta[\log(1 - \mathcal{F}(s + \delta))])$

  Clip$||\delta|| \leq \epsilon$

end for

Stage 2: Steganalysis loss and masking threshold loss both join for optimizing the loss function.

α=0.06

for i = 0 to N_2-1 **do**

  $\delta = \delta - l_{r2} \cdot \nabla_\delta(\mathbb{E}_\delta[\log(1 - \mathcal{F}(s + \delta))] + \alpha \mathcal{L}_{\text{m}}((s + \delta), \delta))$

  if i % 10=0 and $\mathcal{F}(s + \delta)$=1 **then**

    α=α·0.8

  end if

  if i % 30=0 and $\mathcal{F}(s + \delta) \neq 1$ **then**

    α=α·0.3

  end if

end for

Algorithm 1 Two-stage optimization strategy of objective function.

4 Experiment results

4.1 Experiment setup

TIMIT corpus and UME corpus are two widely-used datasets in speech recognition, speaker recognition and so on. Both TIMIT [6] and UME [21] contain uncompressed mono audio with a sampling frequency of 16 kHz. We have conducted experiments on the two datasets to verify the effectiveness of the proposed audio steganography method, respectively. To facilitate the experiment, we have tailored the audio sample to small clips with 16384 sampling points. In the attack process, the embedding rate of LSBM is set to 1 bit per sample (bps), this is to find an optimal perturbation that makes the stego enjoy with excellent undetectability. The trained steganalyzer is implemented on Lin-Net. The mini-batch size is set to 32. The Adam optimizer is employed with learning rate 0.0001. The proposed method is implemented with PyTorch. The source code is available at https://github.com/Chenlang2018/Imperceptible-Adversarial-Audio-Steganography-Based-on-Psychoacoustic-Model.

The hyperparameters $\alpha, l_{r1}, l_{r2}, \epsilon$ are of significant importances to our proposed method. To obtain the four appropriate hyperparameters, we have researched the influence of hyperparameters variation on the perceptual quality as demonstrated in Fig. 3. In order to quantitatively assess the audio perceptual quality, we adopt the widely-used reference audio quality metrics, i.e., the subjective metric PESQ [22] and the objective metric SNR (Signal-to-Noise Ratio). SNR characterizes the average power ratio between the intrinsic signal and the noise. PESQ score ranges from -0.5 to 4.5, and a higher value indicates better perceptual quality. We randomly select 100 test audio samples from TIMTI dataset as references and

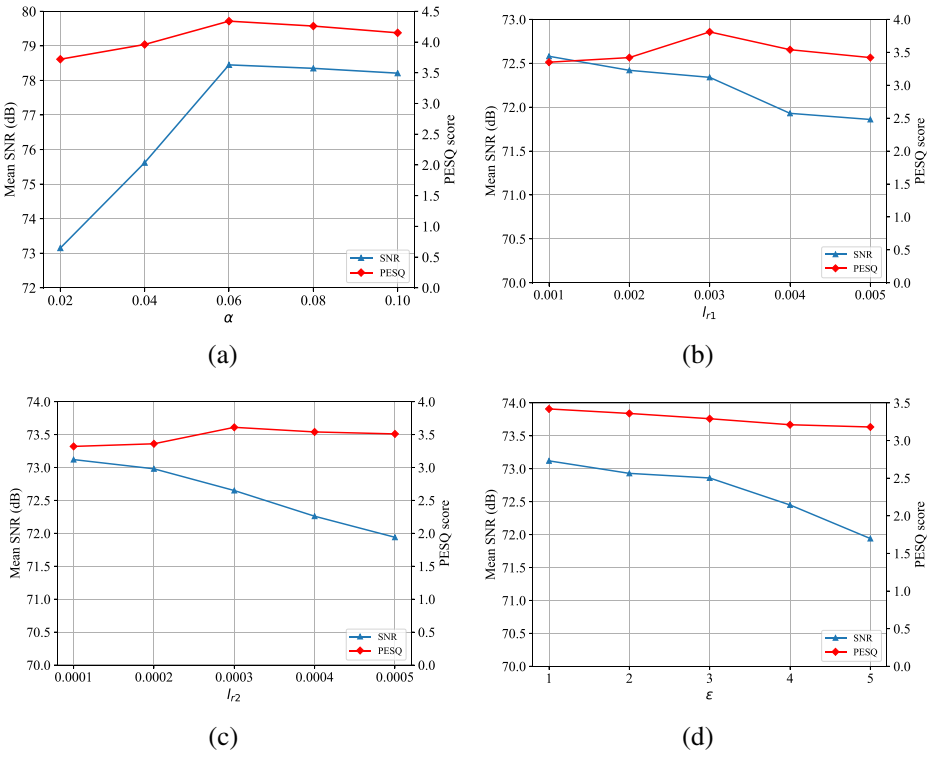


Fig. 3 The Mean SNR and PESQ score evaluated as the four hyperparameters increase, respectively

corresponding adversarial cover audio for evaluation. We can observe that gradual aggrandizement of PESQ with the increase of hyperparameter α, while α increases to 0.06, the PESQ value begin to degrade. The similar phenomenon can also be observed on SNR. To sum up, the proper value of hyperparameter α is 0.06, l_{r1} is 0.003, l_{r2} is 0.0003, ϵ is 1.

4.2 Visualizations

To illustrate the effectiveness of our optimization strategy intuitively, the visualization results of one randomly selected adversarial cover in the two-stage optimization process are shown in Fig. 4. As one can see, the optimized perturbation in stage 1 is quite large in magnitude, this indicates that the adversarial cover audio in stage 1 shares a low perceptual quality. In stage 2, the perturbation in stage 1 is optimized once again according to the masking threshold loss. Finally, the optimal perturbation is so weak that the adversarial cover audio equipped with excellent imperceptibility. The similar phenomenon can be observed in the spectrogram of the adversarial cover audio. To sum up, our proposed two-stage optimization strategy can generate imperceptible perturbation to attack the steganalyzer.

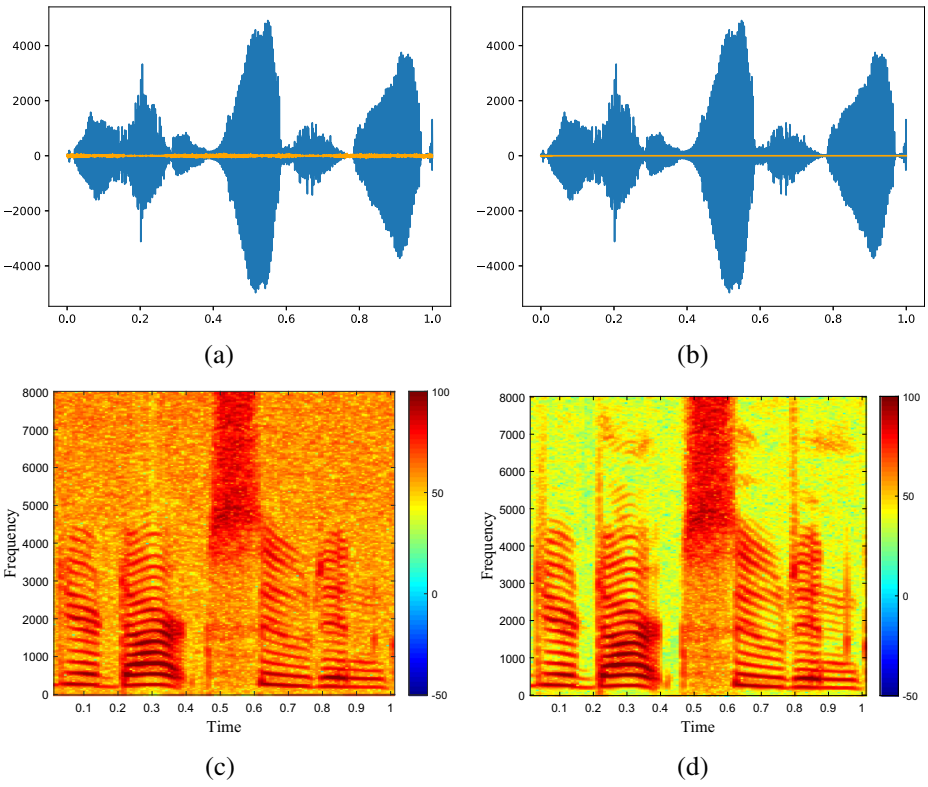


Fig. 4 The visualization results of one randomly selected adversarial cover audio on TIMIT. (a) is the waveform of the adversarial cover audio and corresponding perturbation in stage 1. (b) is the waveform of the adversarial cover audio and corresponding perturbation in stage 2. (c) is the spectrogram of the adversarial cover audio in stage 1. (d) is the spectrogram of the adversarial cover audio in stage 2

4.3 Evaluation of undetectability performance

In this section, we have conducted two experiments on TIMIT and UME dataset. We compare the detection accuracies with several classical audio steganogrpahy schemes, i.e., LSBM [23], STC [4], AAC [17]. It should be remarked that we define the STC distortion cost as 1 to facilitate the experiment. Two state-of-the-art deep learning based audio steganalysis methods in the time domain, i.e., Lin-Net [15] and Chen-Net [3] are used for evaluating the undetectability performance of these steganography schemes. Moreover, the traditional audio steganalysis methods based on handicraft features from audio residuals, i.e., Luo-Residual [16] is also employed to evaluate the undetectability performance. As for the experiments on TIMIT, 15000 audio clips from TIMIT are used to generate the adversarial cover audio samples. Then we embed the bitstream secret messages into the adversarial cover audio samples. This finally forms 15000 cover-stego pairs. 12000 pairs of them are used as training set, and the remaining 3000 pairs are for testing set. We have performed steganography experiments on five embedding rates, i.e., 0.5 bit per sample (bps), 0.4 bps, 0.3 bps, 0.2 bps, and 0.1 bps. For the purpose of reducing the randomness of the experiment results, we repeat all the steganalysis experiments for 10 times under randomly splitting training set and test set and then average the detection accuracies. The similar experiments

Table 1 The detection accuracies (%) of four audio steganography schemes on TIMIT dataset

Steganalysis methods	Embedding rates	Steganography schemes			
		LSBM [23]	STC [4]	AAC [22]	Proposed
Lin-Net [26]	0.1 bps	60.24	52.32	52.45	**49.29**
	0.2 bps	67.45	55.18	56.13	**51.19**
	0.3 bps	69.15	62.35	61.98	**56.85**
	0.4 bps	72.54	68.72	69.14	**57.34**
	0.5 bps	76.28	70.12	70.05	**60.19**
Chen-Net [15]	0.1 bps	59.41	50.19	50.47	**48.17**
	0.2 bps	63.35	54.95	55.49	**51.02**
	0.3 bps	67.23	60.25	60.92	**55.29**
	0.4 bps	70.22	63.48	64.12	**56.29**
	0.5 bps	72.15	68.22	69.03	**58.49**
Luo-Residual [17]	0.1 bps	60.72	51.86	52.09	**47.97**
	0.2 bps	64.18	57.32	58.23	**51.45**
	0.3 bps	67.49	58.54	59.16	**54.39**
	0.4 bps	69.26	59.93	60.03	**55.35**
	0.5 bps	71.36	61.02	61.24	**57.28**

The values in bold denote the lowest detection accuracies

Table 2 The detection accuracies (%) of four audio steganography schemes on UME dataset

Steganalysis methods	Embedding rates	Steganography schemes			
		LSBM [23]	STC [4]	AAC [22]	Proposed
Lin-Net [26]	0.1 bps	60.15	51.13	51.87	**49.25**
	0.2 bps	67.38	56.49	55.69	**51.03**
	0.3 bps	70.24	60.12	60.82	**56.12**
	0.4 bps	72.35	68.27	68.96	**57.29**
	0.5 bps	75.24	71.08	69.81	**60.05**
Chen-Net [15]	0.1 bps	58.31	50.89	50.35	**48.01**
	0.2 bps	60.14	52.65	54.76	**49.89**
	0.3 bps	63.49	59.49	60.34	**54.28**
	0.4 bps	65.21	62.08	63.59	**54.96**
	0.5 bps	71.65	65.21	68.94	**56.11**
Luo-Residual [17]	0.1 bps	60.26	51.25	51.98	**47.85**
	0.2 bps	64.39	57.11	58.02	**51.24**
	0.3 bps	66.87	58.42	58.82	**54.21**
	0.4 bps	69.32	59.16	59.90	**55.18**
	0.5 bps	70.71	60.85	60.89	**57.33**

The values in bold denote the lowest detection accuracies

as stated above are also conducted on UME. The detection accuracies on TIMIT and UME dataset are tabulated in Tables 1 and 2, respectively.

As one can observe, the undetectability performance of the three comparative steganography schemes are relatively poorer than the corresponding results of the proposed steganogrpahy methods under all the five embedding rates. This means that our proposed method can generate better adversarial cover audio for message embedding. With more detailed comparisons, for low embedding rates, e.g., 0.1 bps, the detection accuracies of our proposed are all lower than 50%, this approximate to the randomly guess. While LSBM, STC, and AAC still attain relatively higher detection accuracies. For large embedding rates, e.g. 0.5 bps, the detection accuracies of our proposed method are all inferior to 61% when detected by the three state-of-the-art audio steganalysis methods. This means that our proposed method can defeat both the deep learning based audio steganalysis methods and the traditional audio steganalysis methods. The similar phenomenon as shown can be observed from Table 2. In a nutshell, the undetectability performance distinctively outperforms that of the three classical audio steganography schemes. It is because that our proposed method is devoted to fooling the target steganalyzer in the optimization process, which enables the adversarial cover audio more satisfactory for message embedding.

5 Conclusions

In this work, we proposed an imperceptible audio steganography with psychoacoustic model. In the optimization process, we first add perturbation on the stego audio for constructing noise stego audio that is delivered to the trained steganalyzer for misclassifying. The perturbation is constantly optimized in the adversarial process, this is aimed at seeking for an optimal perturbation that guarantee the imperceptibility and undetectability of stego audio. Empirically speaking, it's difficult to utilize the gradient back-propagation to optimize the threshold loss function directly because of its higher order in magnitude. So we adopt two-stage optimization strategy to minimize the loss function, the first stage aims to find a suitable perturbation to deceive the steganalyzer, the second stage concentrates on further optimizing the perturbation to make the stego imperceptible. For the practical steganography, the optimal perturbation obtained from the adversarial attack process is added on the original cover audio to construct the adversarial cover audio. Then one can use information embedding method (e.g., LSBM) to embed the secret message on the adversarial cover to generate stego audio. It should be remarked that the secret message can be correctly extracted as usual. Extensive experiments show that our proposed method can form the adversarial cover audio with high perceptual quality, and the undetectability performance enhanced 2.62-15.19 percent points compared with the classical audio steganography schemes.

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Data Availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of Interests The authors declare that they have no conflict of interest.

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